Cal Poly Pomona

Benchmarking AES-128 Encryption and Decryption on a Common Architecture

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What is AES?

- **Advanced Encryption Standard (AES)**
  - Symmetric key block cipher
  - Operates on **128-bit data blocks**
  - Key sizes: 128, 192, 256 bits
  - Standardized by **NIST (FIPS 197)**
- **Core Properties:**
  - High security and efficiency
  - Parallelizable transformations
  - Widely used in hardware + software systems

AES Algorithm Structure

- **AES-128 Encryption (10 Rounds)**
- **Each round consists of:**
  - SubBytes (nonlinear substitution using S-box)
  - ShiftRows (row-wise permutation)
  - MixColumns (matrix multiplication in $GF(2^8)$)
  - AddRoundKey (XOR with round key)
- **Decryption:**
  - Applies **inverse operations**
  - Executed in **reverse order**

Real-World AES Usage

- **Where AES is Used:**
- HTTPS (TLS/SSL encryption)
- Secure Wi-Fi (WPA2/WPA3)
- Banking systems & online payments
- Disk encryption (BitLocker, FileVault)
- Embedded systems (IoT, routers, phones)
- **Why AES-128?**
- Lower latency than AES-256
- Strong security with high speed

Project Objective

❖ **Goal of This Work**

- Implement AES-128 encryption + decryption
- Use **same hardware architecture** for both
- Benchmark performance fairly

❖ **Key Metrics:**

- Latency
- Throughput
- Resource utilization
- Power consumption

Architecture Overview

❖ **Unified Hardware Datapath Architecture**

- Single AES core handles:

  Encryption
  Decryption

- Mode controlled via **enc_dec signal (which we set as sw[0])**

❖ **Key Idea:**

- Same datapath → identical conditions
- Enables **true benchmarking comparison**

Design Features

- **Engineering Design Choices**
- Repetitive architecture (1 round per cycle)
- Finite State Machine (FSM) control
- Internal key expansion
- Register-based state storage
- **Benefits:**
- Low hardware overhead
- Deterministic timing
- Scalable design

Why Same Architecture Matters

- **Benchmarking Fairness**
  - Using identical hardware ensures:
  - Same critical path delays
  - Same resource usage
  - Same clock conditions
- **Result:**
  - No bias between encryption vs decryption
  - Direct apples-to-apples comparison

FPGA Implementation

- **Hardware Platform: Nexys A7**
- Developed in **Verilog HDL**
- Synthesized in **Xilinx Vivado**
- Clock frequency: **100 MHz**
- **Implementation Details:**
- No DSP slices used
- Uses LUTs + flip-flops
- Fully synchronous design

Software Processing & Benchmarking

❖ **Why This Design is Good for Benchmarking**

- Controlled FSM execution
- Deterministic cycle count
- Identical datapath reuse

❖ **Processing Characteristics:**

- 1 AES round per clock cycle
- Fixed latency per block
- No pipeline variability

Performance Results

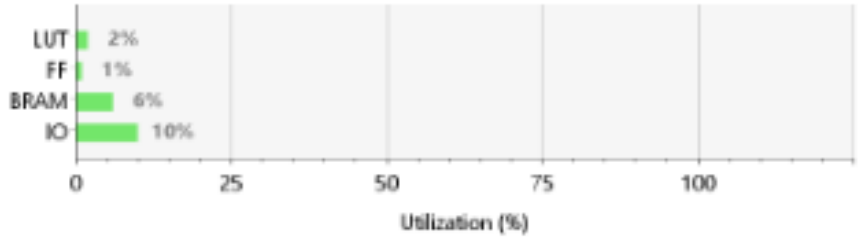
- **Measured Performance**
  - Got Latency through in program Calculation
  - Latency: **10 clock cycles per block**
  - Throughput: **1.28 Gbit/s =** $\frac{Block\ size \times Clock\ Frequency}{Cycles\ per\ block}$
- **Validation:**
  - Verified using AES test vectors
  - Correct ciphertext + plaintext recovery

```
ENC done in 10 cycles
ENC result = 69c4e0d86a7b0430d8cdb78070b4c55a
DEC done in 10 cycles
DEC result = 00112233445566778899aabbccddeeff
```

Resource Utilization

- **Hardware Efficiency**
- Minimal LUT usage
- Low flip-flop count
- No DSP usage
- **Implication:**
- Suitable for **resource-constrained systems**
- Efficient logic-based implementation

Resource	Utilization	Available	Utilization %
LUT	953	63400	1.50
FF	168	126800	0.13
BRAM	8	135	5.93
IO	20	210	9.52

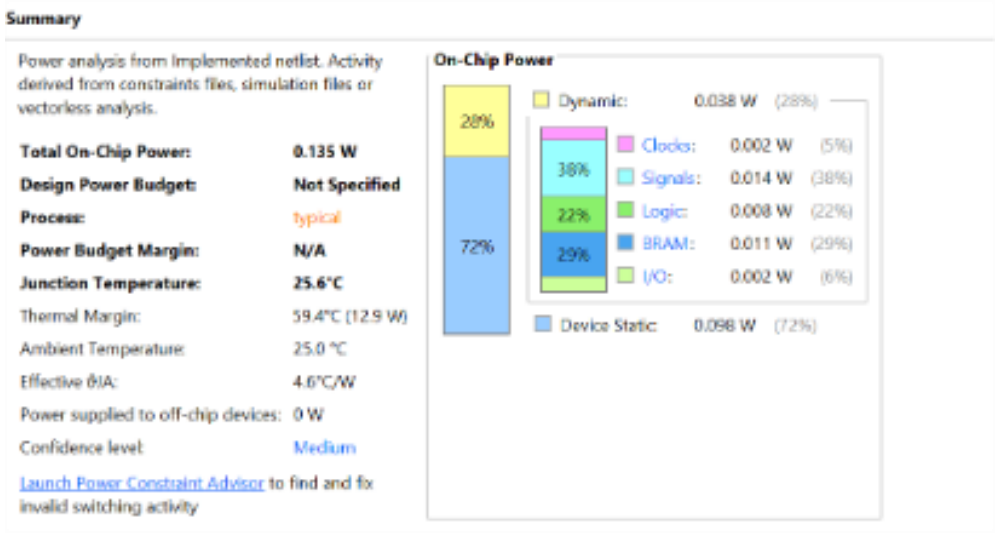


Power Consumption

- **Energy Efficiency**
- Low power usage observed
- Suitable for:

  Embedded systems

  Battery-powered devices



Encryption vs Decryption Comparison

Case	Clock (MHZ)	Cycles/Block	Throughput (Gbit/s)	LUT	FF	BRAM	DSP	Power (W)
AES-128 Encrypt	100	10	1.28	953	168	8	0	0.135
AES-128 Decrypt	100	10	1.28	953	168	8	0	0.135

The program we created has set markers and calculated the mean for both encryption and decryption. This allows for accurate comparison because the result for these two should be close to identical.

Why Results Matter

❖ **Engineering Insight**

- Confirms symmetry of AES algorithm
- Demonstrates fairness of benchmarking method
- Validates unified datapath design that was used

UART Demo Explanation

❖ **UART Interface Demonstration**

- UART = Universal Asynchronous Receiver/Transmitter
- Used for:

  Sending plaintext to FPGA

  Receiving ciphertext output

❖ **Operation Flow:**

- Input plaintext via UART
- FPGA processes AES encryption/decryption
- Output transmitted back via UART

❖ **Engineering Significance:**

- Real-time hardware validation
- Demonstrates system I/O integration

```
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
PLAINTEXT : 00112233445566778899AABBCCDDEEFF
KEY       : 000102030405060708090A0B0C0D0E0F
CIPHERTEXT: 69C4E0D86A7B0430D8CDB78070B4C55A
DECRYPTED : 00112233445566778899AABBCCDDEEFF
```

Real-Life Application of Results

- **How This Applies in Practice**
- Hardware encryption engines in:
  - Routers
  - Secure processors
  - IoT devices
- **Advantages of This Design:**
- High throughput
- Low latency
- Low power
- Minimal hardware footprint

What Was Achieved

❖ **Project Accomplishments:**

- Fully functional AES-128 system
- Unified encryption/decryption architecture
- Verified correctness with test vectors
- Benchmarked performance metrics

Conclusion

❖ **Final Takeaways**

- Unified architecture ensures **fair benchmarking**
- AES performs efficiently on FPGA hardware
- Design achieves:
  High speed
  Low resource usage
  Energy efficiency

❖ **Key Idea:**

- Same architecture for AES → identical performance for AES